

Reducing Voluntary Turnover: A Data-Driven Analysis of Attrition Risk Factors

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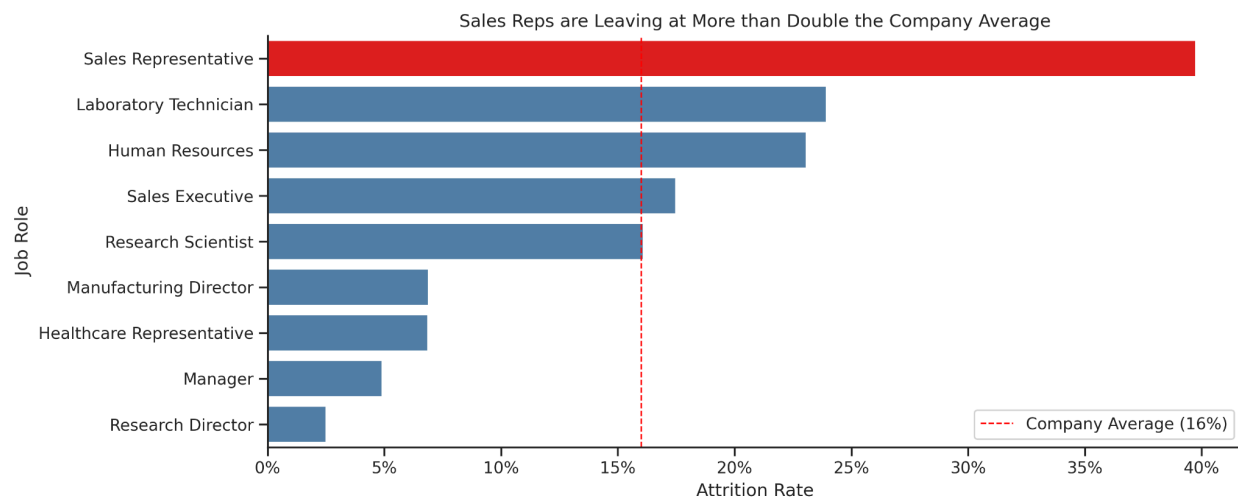
Introduction

Attrition is a major concern for any organization, representing both the loss of experienced staff and the cost of hiring and training new employees. This report addresses the following business question: “Which employee segments have the highest risk of attrition, and what operational and structural changes should HR leadership make to reduce voluntary separation in those roles?” The target audience for this report is HR leadership. The data used in this analysis is the IBM HR Analytics Employee Attrition & Performance Dataset, sourced from Kaggle. This dataset contains 1,470 records across 35 variables including attrition status, role, tenure, overtime, and monthly income.

Data Preparation

The dataset did not require any missing value decisions, all 1,470 records were complete across all 35 variables. Attrition and OverTime, both key categorical variables, were encoded as numeric dummy variables to allow for aggregate calculations by group. Attrition and Overtime rates were calculated as the mean within each group, representing the proportion of each group that had left the company or worked overtime. JobRole was left in its original form for categorical comparisons. EmployeeCount, StandardHours, and Over18 were identified as constants and excluded from analysis.

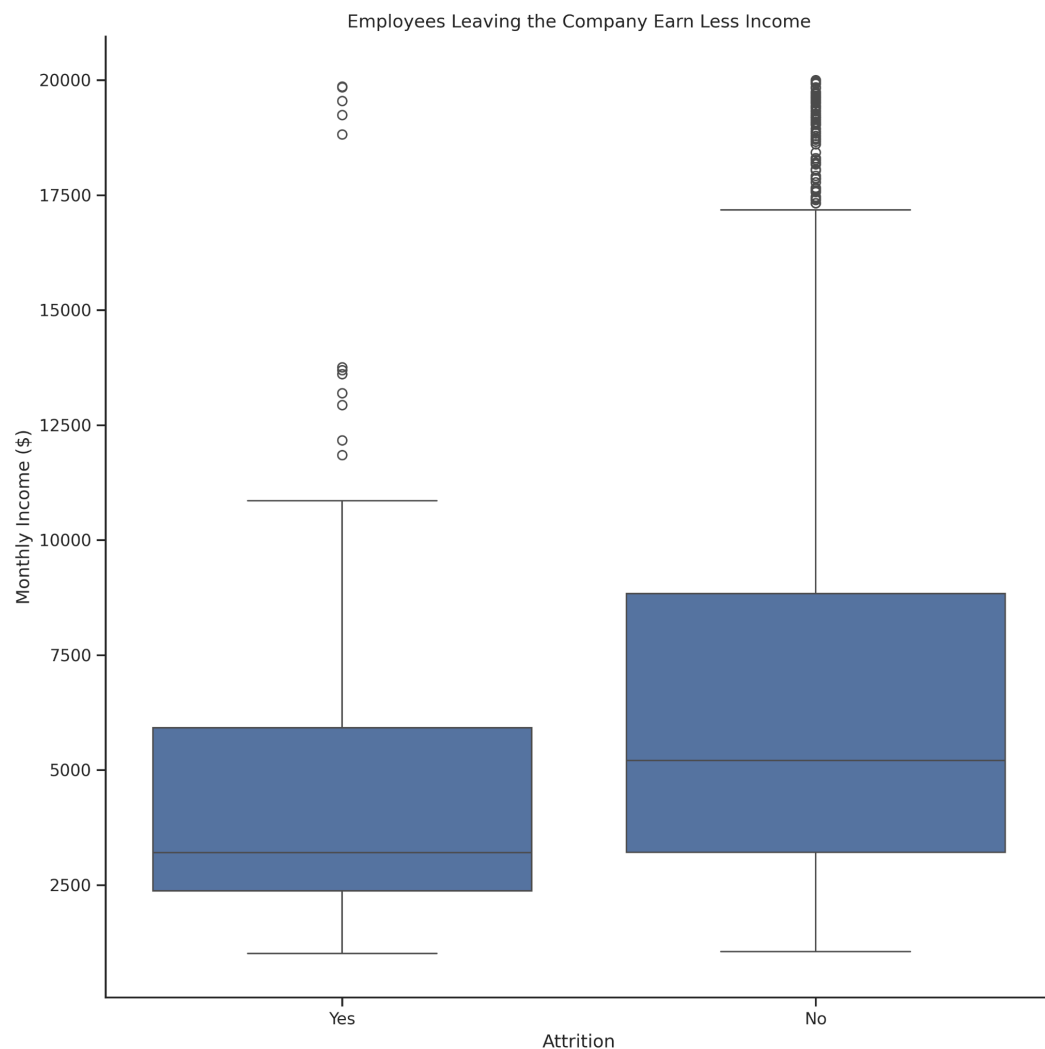
Visualizations, Design Rationale, and Analysis



Attrition rates are not evenly distributed across all roles. Sales Representatives are leaving at significantly higher rates than any other role, nearly three times the company average. Sales Representatives leave the company at a higher rate than the 5 lowest attrition roles combined. Laboratory Technicians and Human Resources also leave the company at a higher than average rate, suggesting there are other factors to be explored.

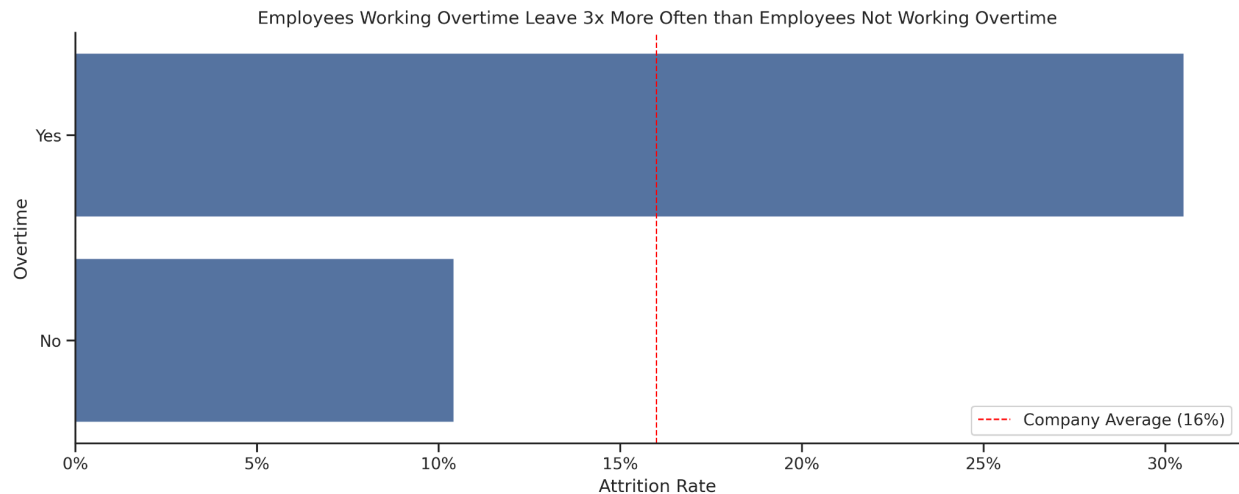
A bar chart was chosen to display rank comparison across nine categories, with a horizontal orientation chosen to accommodate the length of the job titles. Red was chosen as a

preattentive attribute to highlight the significance of the attrition rate for Sales Representatives. Seaborn's despine function was used to remove chart borders to increase data-to-ink ratio. The company average line was included to provide a clear reference for an HR leadership audience.



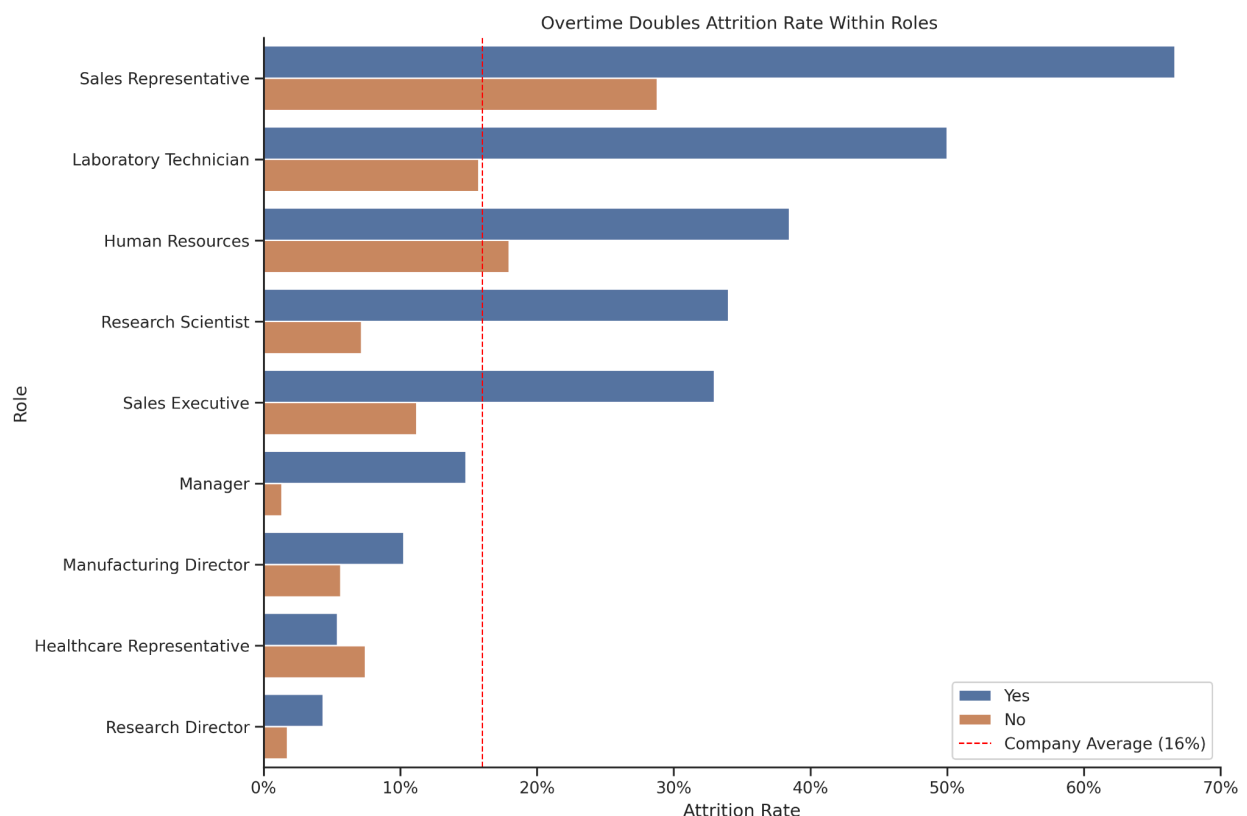
The median income for employees leaving the company is around the 25% percentile of employees who stayed, and the 25% percentile of employees leaving the company is below \$2,500 a month. While there are some outliers who earned significantly more income and still left the company, this indicates that income is a contributing factor in attrition, if not the sole driver. Compensation alone does not explain attrition rates across roles, overtime emerges as another significant factor worth exploring.

A box-and-whisker plot was chosen to highlight the range of income values. A bar chart of median income would have shown the disparity of incomes but not the insight of the median income for attrition being around the 25% percentile of those staying with the company. The (\$) was added in the Y-axis label to provide context to the viewer. Color was kept consistent between the two boxes as the positional difference was sufficient to make the distinction between the groups. Seaborn's 'ticks' theme and despine were used to maximize the data-to-ink ratio.



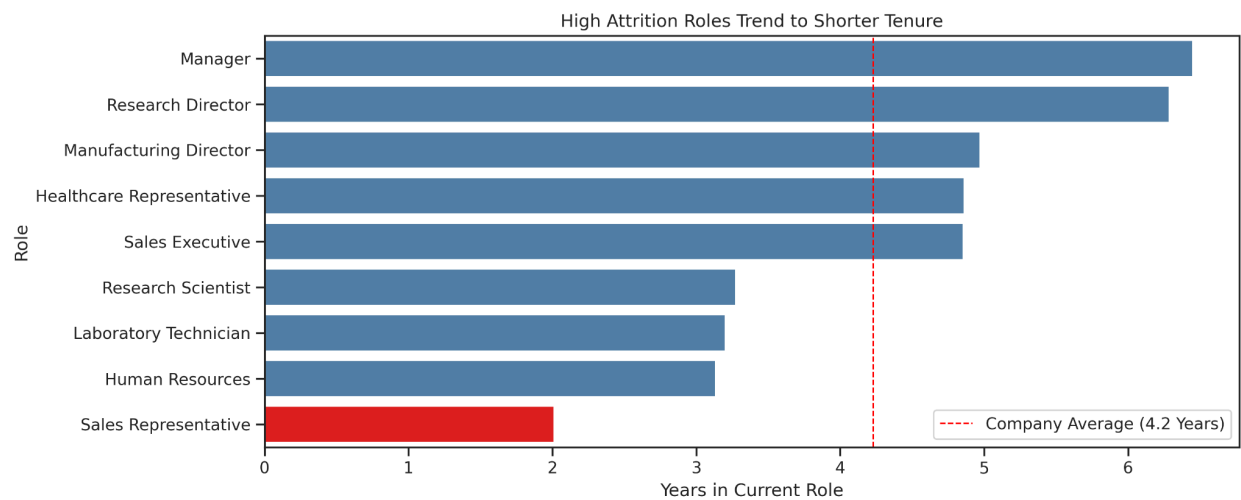
Employees working overtime are twice as likely to leave as the company average, and three times as likely as employees not working overtime. This indicates that working overtime is a significant factor in attrition company-wide. When broken down within roles, this will become even more apparent.

A bar chart was chosen to compare attrition rates between two categorical variables, with the horizontal orientation retained for consistency across charts. Color was kept consistent between the two bars due to the large disparity being easily understood and to keep the chart simple. An analytical title, "Employees working overtime leave 3x more often than employees not working overtime" was chosen to convey the gap further. Seaborn's despine was used to increase the data-to-ink ratio.



Within job roles, employees working overtime across nearly every role are more likely to leave the company, with the single exception of Healthcare Representative. Laboratory Technicians working overtime leave the company at three times the company average, while employees in the same role not working overtime leave below the company average. Sales Representatives working overtime have an attrition rate of 66%, meaning two out of three sales reps working overtime will leave the company. Sales Reps not working overtime still leave the company at nearly double the company average, suggesting additional factors which become clearer when examining tenure by role.

A clustered bar chart was chosen over a stacked bar chart to avoid comparison issues that would arise from the stacked chart not sharing a common baseline. A two-color palette was chosen to distinguish between groups without implying a value judgement. The company average line and percent formatting are included to provide context for the HR leadership audience.



Sales Representatives have the shortest tenure of any role in the company, followed by Human Resources and Laboratory Technician, which represent the three highest attrition roles. The combination of high attrition and short tenure indicates that employees in these roles do not see a future within the company. Across all five visualizations, a consistent pattern emerges: Sales Representatives, Laboratory Technicians, and Human Resources employees face compounding pressures of below-average compensation, high overtime demands, and short tenure in role. Addressing these factors through targeted changes represents the most effective path to reducing attrition.

A bar chart was chosen over a box plot to keep the data readable across all 9 categories. Sales Rep is highlighted in red to show a mirror image of the first chart. The average line is chosen to communicate Sales Representatives average nearly half the company average for tenure to HR leadership. The sort order is ranked highest to lowest and horizontal orientation retained to be consistent with the other charts.

Conclusion

Three recommendations emerge from this analysis. First, Sales Representatives and Laboratory Technicians working overtime are leaving the company at 67% and 50% respectively, much higher than the 16% organization-wide average. HR leadership should audit overtime requirements in these roles, with a focus on reducing non-essential overtime. HR leadership should evaluate if the benefit of these employees working overtime is greater than the cost of hiring and training new employees for these roles. Second, Sales Representatives have the highest attrition rates and the lowest time in Role, suggesting they don't see a future at the organization. Providing a clear promotion pathway or lateral moves within the organization could ensure they're pursuing opportunities internally rather than externally. Finally, employees leaving the organization are earning significantly less income: the median income of employees who leave the organization is around the 25% percentile of those who stay. A review of compensation for high attrition roles is warranted based on this data. However, there are outliers who earn a high income and leave the company, meaning compensation is not the answer on its own. This dataset does have its limitations, it was created for a data analysis project such as this and does not contain actual data for a real company. Due to its cross-sectional nature, causality cannot be established, only correlation. Further investigation could be conducted on additional factors concerning Sales Representative attrition, employees not working overtime are still leaving at a 29% rate, implying there are more factors at play than are contained in this analysis.

References

Kaggle Dataset:

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset/code>